

Behind the failure of Theranos

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Introduction

Theranos is a private medical technology company based in California and founded by Elizabeth Holmes in 2003 (Rago). The company aimed to revolutionize the laboratory blood-testing industry with its "Edison" machines. Instead of needing multiple blood vials and big testing machines, the "Edison" would break the industry standard, requiring only droplets of blood and one table-size testing machine (Rago).

By February of 2015, with more than 400 million U.S dollar investment, Theranos was valued at 9 billion U.S dollars (Austin et al.). Theranos's CEO, Elizabeth Holmes, rose to celebrity status and appeared on the cover of multiple renowned business magazines like Forbes and Fortune (Kulwin). In addition, the White House even made her the U.S Ambassador of Global Entrepreneurship (Rao).

Things were going great for Theranos until October 2015, when John Carreyrou of the Wall Street Journal published his article: "Hot Startup Theranos Has Struggled with Its Blood-Test Technology.". With scientific findings and statements from former employees, the feature exposed the lie behind the scene that none of Theranos' machines work. John Carreyrou's work also pushed for further investigation, which revealed the manipulation and falsified statements.

With the endless amount of lawsuits and scandals, Theranos officially dissolved in 2018 after 15 years of existence (Carreyrou). As of 2022, Theranos' CEO, Elizabeth Holmes, continues to face legal trials for fraud and conspiracy. She may expect up to 20 years in prison (Paul).

Why study Theranos?

The Lost Opportunity

Despite the controversy around Theranos, people can't repudiate that the company secured more than 400 million dollars of investment (Austin et al.). For her board of directors, Elizabeth Holmes did manage to convince influential political figures like Henry Kissinger, former U.S Secretary of Defense, or George Pratt Shultz, former U.S Secretary of the Treasury (Pflanzer). Moreover, she recruited renowned tech figures like Avie Tevanian, former chief software technology officer at Apple (Dunn et al.). Elizabeth Holmes has all the resources and talents. She and her company could have, for sure, revolutionized the medical industry if they didn't so desperately chase an impractical and delusional vision. Likewise, maybe Therano's secret could have stayed in the dark if Elizabeth Holmes didn't create such a toxic working environment for her employees. In the end, Theranos is more than just a fraudulent company. It is a waste of huge potential/opportunity.

The Implication

The case of Theranos altered the public preconception about medical startups. Before 2015, Theranos was the center of attention, drawing in all funding and investment and away from other legitimate medical startups (TheVerge). After 2015, Theranos remain haunting the industry, creating a stigma and skepticism against those same legitimate medical startups (Mak). Is the customary Silicon Valley motto "Move fast and break things" the best thing to apply to medical startups when people's health and lives are at risk (TheVerge)? The answer is unclear for the time being. Yet, Theranos for sure has delayed or prevented any progress for humanity to acquire better healthcare (Mak).

Applying Management & Organizations

This research paper will analyze the entity that was Theranos through multiple course concepts from “MGMT-SHU 301” (specifically leadership and organizational change, organizational culture, and organizational strategy) for greater in-depth understanding. It will point out what Theranos did right or wrong from the current outcome. It will also give alternative recommendations on what Theranos could have done instead for better results. Hopefully, by the end of this paper, future companies can learn how to achieve the level of power Theranos has and avoid its mistakes.

Research Methods

For research information and resources, we first visited the company's website to obtain basic background content. Second, we reviewed a variety of media materials, including podcasts, news articles and reports, documentaries, and interviews from credible sources such as ABC News and the Wall Street Journal. Third, we consulted business literature and journals.

Culture Analysis

Organizational Culture is a collection of "values, attitudes, beliefs, and expected behavior shared by members of an organization" (Ro). Researchers can study a company's culture by carefully observing factors like competitiveness, individual initiative, innovativeness, tolerance of failure, conflict tolerance, and power distances (Ro). It directly correlates with employees' behavior, individuality, motivation, and commitment (Ro). Altogether, good organizational culture usually means better company output and performance. One example is the case of Microsoft and its current CEO, Satya Nadella. Since taking up the mantle in 2014, Nadella revamping his company's culture, promoting self-improvement and cooperation instead of competition. With the innovational change, Microsoft's market value has risen from 300 billion U.S dollars to over 1 trillion U.S dollars (Business Insider).

On the surface, Theranos should have a good company culture based on astonishing growth and value, but, in reality, it couldn't be furthest from the truth. With regards to competitiveness, employees Elizabeth Holmes divided employees into teams. Then she put these teams head to head against each other in a "survival of the fittest" style. Regarding individual initiative, employees have no freedom and independence. Theranos' former design director, Justin Maxwell, noted everyone was constantly micromanaged and surveilled (ABC News). There is also zero encouragement for innovation at Theranos, shown through factors like the office's layout. Ana Arriola, former Theranos's Chief Design Architect, described the company as promoting "multiple levels of secrecy and compartmentalization " (ABC News). Another former employee says the infamous CEO intentionally separated the department to hide that their machines weren't working (ABC News). Elizabeth Holmes's plan worked so well that only by

the time of Theranos's trial that two co-lab directors know of each other existence (Chang). If employees have questions or concerns about their job, they should remain silent, as Holmes would fire or threaten anyone who asks too much. Even after leaving the company, Theranos would go the extra mile to shush former employees with warning letters and lawsuits (Cheung).

In conclusion, when describing the organizational culture at Theranos, the researchers can only use one word, fear (Chang). The employees were under abnormal mental pressure and stress. The situation was so unpleasant that Ian Gibbons, chief scientist of Theranos, attempted suicide and passed away in 2013 (CBS Mornings). The employees have no trust and loyalty for Theranos the same way the company has none for them. If there is a lesson from all of the above, it would be: that the culture at Theranos is terrible. Future companies and startups need to avoid creating cultures that resemble Theranos'.

Alternative recommendations

In a hypothetical reality where Elizabeth Holmes understands her wrongdoing and wants to improve her organizational culture, what can she do? She would need to look at the study "Duane Morris: Balancing Growth and Culture at a Law Firm" by Boris Groysberg and Robin Abraham. The study from two Harvard professors concludes that Duane Morris's key to success is the Quaker Beliefs, which include equality of all people, simplicity and functionality, and progress through incremental change. Concerning equality for all people, Quakers trust that everyone is equal and that "leadership was a service function." It meant all members of an organization deserve the same level of respect and treatment regardless of their titles. Quakers carried out this first belief with "under-compensation of top management, multitiered compensation practices that rewarded many different kinds of contributions, and an emphasis on diversity and family-friendly policies". Applying the first belief to Theranos, Elizabeth Holmes

would need to respect her employees more by removing the competition between teams, addressing their questions and concerns, and lessening micromanaging and surveillance. The Theranos CEO can also implement a rewarding system to recognize her employees' contributions and teamwork. By carrying out this suggestion, Theranos would return and improve a safe work environment for its employees. Concerning simplicity and functionality, Quakers believe "that any action organization makes need to have a solid functional justification". Applying this belief to Theranos, the company need to change its current office layout as employees have complained about how the office layout damage the information and workflow between department. Concerning progress through incremental change, Quakers believe slow and observed growth helps secure stability and avoid risks. These risks can range from sexual harassment to overworking culture at companies that the public has seen too much in the recent news. These risks entail lawsuits and public relations damage. Applying the third belief to Theranos, Holmes should set up a five-year growth plan. This plan needs to include realistic goals that are relatable to the employees. For example, one of the goals can be to increase company efficiency and shorten working hours. Similarly, one other goal can be: to decrease the number of resignations annually. Overall, a combination of changes based on the Quaker beliefs suggested above would have headed Theranos in a better direction.

Limitations of alternative recommendations

The success of an organization lies in multiple factors working in unison with one another. The suggested recommendation assumes that the organizational culture at Theranos is not restricted by other factors- Elizabeth Holmes carried out this culture change herself. Removing Elizabeth Holmes from the equation or adding another factor may affect the effectiveness of the suggested recommendations. Similarly, the Quaker's Beliefs are logical but

may not be effective on Theranos, a medical tech company, as it did with Duane Morris, a law firm.

Leadership Analysis

A leader determines an organization's duty and vision and how to reach them (Ro). Facing challenges, a leader is also responsible for supporting the organization's duty and vision through clear communication, motivation, and adaptability (Ro). In a similar manner, employees use their leader to set up behavior standards within the organization (Ro). Using this description to evaluate Elizabeth Holmes, was she a bad leader? The answer is no, but she did something right.

Elizabeth Holmes did one thing right as a leader. She set out a vision that speaks to everyone- "a world in which no one ever has to say goodbye too soon." The way Holmes presented the company's vision was breathtaking. It helps the organization recruit its first wave of employees and differentiate the company from competitors.

Elizabeth Holmes was a bad leader for multiple reasons. First, she was a liar. Holmes lied to her employees, her investor, and the public about the functionality of the Theranos. She lied about the tones of her voice. Elizabeth Holmes intentionally lowered her voice tone, believing it would earn more respect from people (Business Insider). Lying became so natural that she lied to her employees about the smallest detail. One former employee recalls her lying about being out of the office but can be seen right down the hall (ABC News). Second, she did not provide support for her organization to achieve the company's vision. Elizabeth Holmes chose to pretend her company had reached its vision already. As pointed out from evidence from analyses of organizational culture above, she actively sabotaged her employees' work, factors that would

contradict the false reality she let herself drown in. Therefore, Her detachment from reality means zero communication, motivation, and adaptability-things that matter.

Alternative recommendations

To address Theranos' problems with leadership, the organization should replace Elizabeth Holmes with a new CEO, or Holmes needs to change her way. Since a former member of the Theranos' board did try and failed to remove Elizabeth Holmes from the CEO position (ABC News), it is more suitable for this research paper to look at the former option. Elizabeth Holmes needs to set her interests-being rich and famous, aside. She needs to build trust with her employees by being honest and truthful with her employees with a normal voice tone. In addition, she should be more practical with her visions and set short-term and long-term goals. Holmes increasing Emotional intelligence would also be a good approach. Being able to sympathize with her employees' feelings, the CEO can notice feedback and further enhance the company's short-term and long-term plans. It is important to note that Elizabeth Holmes's original vision for the company should leave the same as it is beneficial for the company and wasn't part of the leadership problem.

Limitations of alternative recommendations

The success of an organization lies in multiple factors working in unison with one another. The suggested recommendation assumes that the leadership at Theranos is not restricted by other factors. Furthermore, the research paper presumes all portrayals of Elizabeth Holmes's personality and role at the company from the new articles are accurate and covered. Therefore, additional information or changes in the future may make the suggested recommendations here ineffective.

Strategy Analysis

Theranos' strategy also served unrealistic ambitions, a misguided strategy full of ignorance and hypocrisy that led it step by step to failure. As a new laboratory company, Theranos' ambitious vision was to transform healthcare through early detection by using a small number of blood samples to detect multiple diseases. It is easy to see that Theranos' core competencies that distinguish it from its rivals are science and technology and that it should choose to focus on scientific facts rather than inventing myths to deceive the public.

The company's first misleading strategy was to downplay the science. In the company's early years, Ms. Holmes hired more than 80 scientists with Ph. D.s to work on technology development. In 2013, the company also updated a test facility called Edison (Carreyrou). SWOT analysis can help companies to develop and adjust their strategies. It includes a careful assessment of strengths, weaknesses, opportunities, and threats that affect organizational performance. According to the SWOT analysis, high-quality people and advanced equipment were the strengths of this technology-backed company. Theranos, however, ignored the important value of these strengths and thus developed the wrong strategy. According to the complaint filed by the employee, laboratory employees later told federal authorities that the results of quality control runs deviated by more than two standard deviations from known quantities, indicating that there may be problems with the accuracy of the tests. When the Theranos employee reported the matter to his superiors, research and development personnel came to the lab and removed the quality control data to make Edison's test runs look better (Carreyrou). Indeed, scientific experiments require a process. From collection and testing to reporting, clinical, etc., the process must be supported by information and data, and iteratively improved and refined as errors occur. However, Theranos found the enforcement discretion and

regulatory flexibility of the (The Food and Drug Administration) FDA to “use trade secrecy and its corporate dual roles, as both manufacturers of the test and owner of the clinical laboratory, to classify its proprietary machine (which is called “Edison”) as an LDT” (Rohan). In this way, the company appears to have avoided any information from authorities and partners regarding the clinical validation data used to evaluate its tests. The result was that when Theranos' technology began to be falsified and its accuracy could not be certified, its competitiveness faded.

The second faulty strategy was to whitewash reality. By ignoring its core competencies Theranos dug a bottomless pit of lies for itself. To maintain its legend in the marketplace and continue to attract drugmakers and investors, the company had to hide the truth from the public, recalls Annette Campbell-White, managing partner of MedVenture, who says Ms. Holmes was fast-talking and recommended that they put blood tests on the chip. She said Ms. Holmes "talked at 30,000 feet about how these tests would change the human race," but did not provide any technical or scientific details(Carreyrou). To attract investors, Theranos created a report that included the logos of pharmaceutical companies to show that each company endorsed Theranos' technology. In fact, Theranos used the logos without authorization, and the pharmaceutical companies never validated Theranos' technology (Cohen et al.). According to Porter, the five forces that shape strategy are supplier power, buyer power, competitive rivals, threat of substitution, and threat of new entry. Theranos falsified reports in exchange for trust in order to win the buyer's market, which led to a loss of voice in the game with upstream suppliers instead of weakening the buyer's power when the scam was revealed. It also increased the threat of substitutes capturing market share. As we can see, in order to cover up a lie, Theranos had to compile a network of lies. But once one of the lies is uncovered, it leads to an imbalance in all five of its forces.

Alternative recommendations

Theranos' strategic imperative was to get its attitude right. Internally, this was reflected in a pragmatic approach to developing science and technology. Traditional laboratory tools were obtained from other firms by Theranos to test the majority of patient samples, and this data was utilized to compensate for Edison's faults. In fact, Theranos could spend the same amount of time growing its core competency by using information from competitors as materials to train Edison and improve its ability to test. When Theranos' competitiveness is enhanced, Holmes' ambition will have a solid foundation.

For the outside, Theranos is expected to eliminate skepticism. Grand rhetoric is inherently susceptible to questions and concerns, and legends may attract the interest of investors and consumers at the beginning, but over time, when people begin to pay more attention to the internal logic, the company needs to have the real ability to respond, and for Theranos, the ability to respond comes precisely from the improvement of internal technology, with reliable, quality products and services to eliminate the public's doubts. At the same time, it should also take the initiative to accept the testing and certification of authoritative institutions, and make the data information public in a timely manner. Otherwise, doubts will only grow, and the former glory will eventually be short-lived.

Limitations of alternative recommendations

The limitation is that behind the strategy is the vision, and the vision is set by the leaders, so the character and pursuit of the senior leadership represented by Holmes can even be the key to whether Theranos can turn around. But people are the hardest to control, and pinning hopes on individuals carries a great deal of uncertainty and risk.

Closing Remark

Does the Theranos story really serve as a cautionary tale? One day Holmes was being charged, the next day the Biden administration announced a plan to buy 280 million cheap, fast COVID-19 test kits. With the ongoing tension between the speed and reliability of medical testing, how do we balance the risks of disruptive innovation with the deadly costs of administrative inertia (Mazer)? All these questions deserve further thought and practice.

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